

Diversity and Abundance of Macrobenthos between Salimpur Coast and Shapuree Island, Bangladesh

Abstract

Macrobenthic communities are widely recognized as indicators of environmental quality due to their ecological roles and sensitivity to habitat changes. Understanding their distribution and composition is essential for monitoring coastal ecosystem health and guiding sustainable management practices. To contribute to this understanding, this study evaluated the diversity and abundance of macrobenthos in two intertidal regions—Salimpur Coast and Shapuree Island in south east coastal region of Bangladesh. The objective was to identify macrobenthic taxonomy and compare their diversity and abundance across six selected stations—three in each location. Field sampling was conducted in October 2023 during low tide using hand spoon corers. Samples were sieved (0.5 mm mesh), preserved, and analyzed under a stereomicroscope for taxonomic identification. Diversity indices such as Shannon-Wiener were calculated to evaluate species richness and evenness. A total of 148 individuals were recorded, representing 39 families and 15 classes. Annelida was the dominant phylum (84.46%), with Polychaeta accounting for the highest class-level abundance (55.41%). The Nereididae family appeared in all stations except Salimpur Station 2. Salimpur Station 1 exhibited the highest Shannon diversity index (2.22), while Shapuree Station 1 had the lowest (1.62). Shapuree Island recorded higher overall species richness (82 species) compared to Salimpur Coast (66 species), suggesting variability in habitat conditions and substrate type. This study provides valuable baseline data on macrobenthic communities in the selected coastal zones. The results enhance our understanding of species distribution patterns and support future ecological assessments and conservation planning at the local level.

Keywords: Macrobenthos, Diversity, Abundance, Salimpur Coast, Shapuree Island, Bangladesh

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1.Introduction

Macrobenthos in aquatic ecosystems are widely acknowledged as superior bio-indicators for efficiently assessing and monitoring environmental conditions compared to the physico-chemical factors of water and sediment [1]. A group of marine invertebrate organisms known as soft bottom macrobenthos are those that reside in or on the sediment (epifauna) and have direct contact with it [2]. During their life cycle, benthic animals reside within the sediment at the bottom of water bodies or move freely across rocks, organic debris, and submerged portions of plants [3]. There are different types of macrobenthos, and they are divided into classes, families, phyla, and others. For example, Spionidae, Trochochaetidae, Hesionidae, Nereididae, etc.; Nidide Magellanidae, Glyceridae, Phyllodocidae, which is a family of macrobeths. There are several classes of macrobenthos, including polychaete Malacostraca, Clitella, Bivalvia turbellaria, Chromadoria pleistoannelida, and others.

Macrobenthos occupy this ecological niche for the purposes of obtaining food, finding shelter, and facilitating reproduction . Fish and other higher organisms primarily consumed macrobenthos, which served as a crucial component within the food web [4]. The environment, including sediment and water quality, along with hydrological factors influencing the physical habitat, plays a substantial role in shaping the community of benthic organisms. Macrobenthos serves as a biological indicator, reflecting the overall condition of the aquatic environment due to its dependence on the surrounding conditions. [5]. Moreover, factors such as latitude, depth, water temperature, salinity, locally defined parameters like substrate type, and ecological factors like competition and predation collectively impact the diversity and abundance of macrobenthos. The species composition, diversity, productivity, and physiological conditions of local populations in a body of water are all shaped by the interplay of its physical, chemical, and biological characteristics. [6].

The abundance of macrobenthos communities is considered an indicator of pollution in the bodies of water they inhabit. Consequently, their populations vary in response to shifts in physiochemical conditions and the availability of suitable habitats. [7]. Numerous studies have utilized soft-bottom macro-invertebrates as indicators to assess pollution, examining the response of invertebrate communities to alterations in physicochemical factors and available habitats on a global scale [8].

In the study area, Salimpur Coast, and Shapuree Island, samples were collected from 6 stations, with 3 stations in each area. The sediments of the Salimpur coastal zone were muddy and sandy with mangrove grass. The first sampling station was Salimpur coastal beachside area, with sandy sediments, and the other 2 station areas had sandy and muddy sediments. On the other hand, Shapuree Island, which had three stations and was sandy and muddy.

1.1. Objectives

- To collect the macrobenthos from the Salampur coast and Shapuree island.
- To identify the macrobenthos according to family and class.
- To find out macrobenthos abundance and diversity between the Salimpur coast and Shapuree island areas stations.

2. Literature Review

[9] This study dealt with macrobenthos abundance during the post-monsoon period. They found 628 species belonging to 10 major taxa. They showed a difference between the abundance of upstream and downstream macrobenthos, where they found the highest number of macrobenthos downstream.

[10] This paper mainly focused on benthos abundance at 9 stations on the Chittagong coast and 4 stations, namely Parki Beach, Dakshin Kattali, Sitakundu, and Borukundu, from August to September 2008. They identified the macrobenthos according to their taxa, where Amphipoda emerged as the most abundant at Parki Beach and Barubkundu, while bivalves dominated at South Kattli and Seetakundu. The study highlights the high abundance of macrobenthos in the southern Kattli, potentially linked to the elevated organic content in the sediments.

[11] They collected samples from the Tawi River in India. They identified the benthos according to their three main phyla, including Annelida, Arthropoda, Mollusca, and 13 taxa, and they did their work for about a year, from September 2011 to August 2012. They found the maximum number of benthos in the Arthropoda group, which was 57.24%, and they used the Shannon index, Margalef's index, and Menhinick's index for diversity, evenness, and richness. They found the highest number of species diversity in August 2012 and the lowest in July.

[12] They showed the abundance of benthos and aquatic pollution levels in the Maury River, Khulna, Bangladesh, from January to March 2005. They identified species by major taxa in which the Polychaeta group was most abundant, and they worked at 6 stations. 1 was the highest abundance number of benthos, and 1 was the lowest number of benthos, according to the Shannon index.

[13] They studied macroinvertebrate fauna during post-monsoon and pre-monsoon periods in the Kola reservoir of central India. They identified species according to three phyla, including Arthropoda, Mollusca, and Annelida, and found 43 species. They found high variation in the monsoon season according to seasonal variation analysis.

[14] They investigated the abundance, composition, and distribution of macrobenthos in Sungai Pulai Estuary, Johor, Malaysia. They selected seagrass and non-seagrass areas and sediment-dwelling epifauna and infauna. They identified 21 species and polychaete families by family. Population indices such as diversity, richness, evenness, and dominance were employed to assess differences in macrobenthos diversity and abundance across stations.

[7] They collected samples in two seasons, including the dry season and the wet season, to study the diversity of soft-bottom macrobenthos in the Feni Eastuary of Bangladesh. They identified species by family and found 17 families, and Polychaeta was the most dominant species. They used various biodiversity indices, such as Shannon-Wiener index, Margalef's index of species richness, and Shannon equitability index, to assess the diversity and evenness of the macrobenthos community in different sampling stations during the study period.

[15] This paper investigated the presence and abundance of macrobenthos in the Hatiya and Nijhum islands of Bangladesh during the pre-monsoon season of January–June 2010. They collected samples from 7 stations in 2 areas, and these areas were mangrove mudflats, tidal creeks, and reed beds. Species were identified by groups, and 10 groups of species were found. Polychaetes, oligochaetes, shrimp larvae, and crabs were the major species and accounted for 85.24% of the total population.

[16] They mainly studied macrozobenthos in the Krueng Cut estuary, located in Banda Aceh, Indonesia, on 7 stations. They used a ponar grab for collecting samples and sieving them for further analysis. They showed a relationship between macrozobenthos and temperature, salinity, salinity and dissolved oxygen. per square meter they found 347 individuals species in their study area.

[17] They focused on investigating the distribution, abundance, and seasonality of benthic macroinvertebrates in the sediments of Okpoka Creek in the Niger Delta, Nigeria. The research covered a one-year period from January to December 2006. They were identified species by group in six stations. Nineteen species were encountered during the study, with all species of the class Polychaeta, except *Nereis virens*, occurring in all stations. The Shannon-Weaver-Weaver and Margalef index were used to assess diversity, with high values indicating greater species richness. Pielou's index of evenness revealed that species were evenly distributed in the study area during certain months.

[18] They focused on the seasonal and spatial distribution of macrobenthos in the Naf River Estuary of Bangladesh. They identified 47 major taxa and nine groups of species. Polychaeta was the most dominant group, at 60%, and they found a higher mean abundance in the pre-monsoon, and the upper and lower estuaries got the minimum number of abundances. Significant differences were observed in density, the number of taxa, and Margalef's Index among stations and seasons.

[19] They mainly studied the abundance of the polychaete group benthos on the South Indian coast in 2010. They identified 63 types of polychaete group species, with 38 from the West Coast and 25 from the East Coast. Their final results show that sand or sandy sediment areas are the most suitable habitat for polychaetes.

[20] The authors showed the abundance and families of the Macroinvertebrata family in the Ganges River in 2022. They analyzed 15 pollution sites in 4 sequences at 33 locations. They showed the relationship between macroinvertebrates and physicochemical factors (ph, temperature, salinity, and dissolved oxygen).

[21] They showed correlations with abundance and physicochemical indices of macrobenthos in Funka Bay, Japan, during 2001–2004 and 2007–2010. They identified species by three main groups, including long-lived species, sedentary polychaeta (*Notomastus laterius*), and short-lived species. Results showed that where dissolved oxygen on the ocean floor decreased, macrobenthos had higher abundances in 2007–2010 than in 2001–2004.

[22] This paper showed unexpected macrobenthos abundance in the southeastern Chukchi Sea. They found a predominant number of bivalves in *Macoma calcaria*. They showed the results of benthos food sources, and *Macoma* was the most suitable for their study area.

[23] The study showed the biodiversity and abundance of macrobenthic invertebrates in the Datta-Talab pond, located in Jammu, India. They identified 14 species belonging to 3 major taxa, including Annelida arthropoda mollusca. Arthropoda was the most dominant group, accounting for 43.19%. The calculated indices, including Simpson's Index, Shannon-Weiner Index, Maraglef's Index, Menhinick's Index, and evenness, provide insights into species diversity, richness, and community structure.

3. Materials and Methods

3.1. Study Area:

The samples were collected from six stations where 3 stations of Slimpur coast and three stations of Shapuree Island.

3.1.1. Salimpur Coast

The study location is 15 km from Chittagong port city and is situated on the Salimpur coast in Chittagong. [24]. Salimpur coast is an artificial mangrove area which is the northern part of the city. After low tide the area is thick mud and heavy. We collected samples after low tide between 9 am and 1 pm on October 14th.

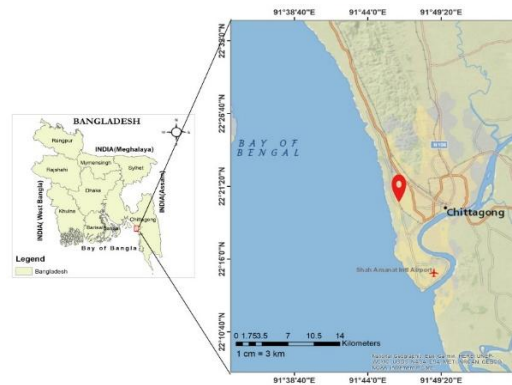


Figure 1: Study area and Map of Salimpur Coast

3.1.2. Shapuree Island:

The Bangladeshi area of Shapuree is situated near the mouth of the Naf River, which serves as a maritime border between Bangladesh and Myanmar. This area was muddy and sandy, and samples were collected after low tide. According to sample collection, this area was divided into 3 stations, which were stations 1, 2, and 3. The first station was close to the beach, and all stations had the same area for sampling.

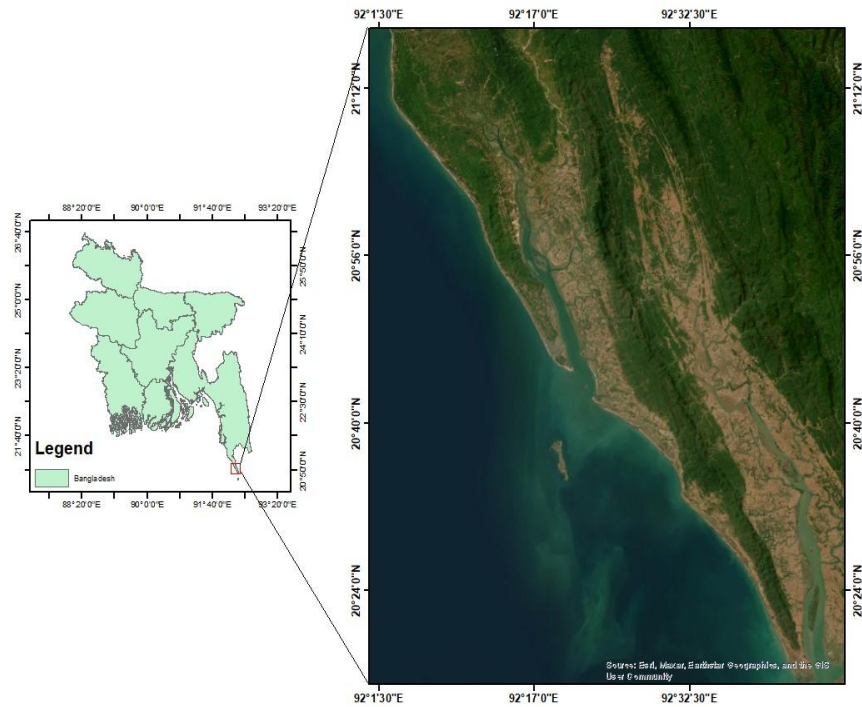


Figure 2 : Study area Map of Shapuree Island.

3.1.3. Stations coordinate and sampling time:

Stations	Latitude	Longitude	Time	Date
Salimpur Coast	22° 23' 33" N	91° 45' 53" E	9 AM – 1 PM	14 October
Shapuree Island	20.7683° N	92.3097° E	2 PM – 5 PM	15 October

3.2 Sample Preparation:

The sediment samples were collected from sampling area during low tide and using 10cm hand corer and 0.5mm hand sized mesh sieve . Samples were collected from the intertidal zone and a 1 m square plot was created for benthos collection. After 1 m distance, again 1 m square was made, similarly 3 stations from Salimpur and 3 stations from Shapuree Island, total 6 stations were used for sampling. The sampler was inserted into the sediment, and the collected sediment from the corer was placed in a polyethylene bag with proper labeling. Subsequently, these sediment samples were moved from the polyethylene bag to a bucket, where they were mixed with water. The mixture underwent filtration through a 0.5 mm mesh-sized hand-sieve to eliminate waste particles. The separated macro-benthos, along with other residues, were then preserved in a labeled plastic container using a 10% buffered formalin solution. Finally, these preserved samples were transported to the laboratory for further analysis.



Figure 3 : Samples collections from Study area of Salimpur coast and Shapuree Island.

3.3. Sample Analysis

In order to make identification easier, the samples were placed in 15 cm diameter by 2 cm deep, round, transparent Petri dishes. To improve the contrast in appearance, these plates were set on black tissue paper. Using a small brush or forceps, the organisms in the samples were sorted and counted according to main family, class and the recognized specimens were then kept in tiny bottles. A digital camera (Model No. Optica C-B5+) attached to a binocular microscope was used to identify and take pictures of benthos. After identifications samples were preserved in 70% ethanol solution.



Figure 4 : Species identification at lab

3.4. Statistical Analysis

Statistical analysis of benthos was done on their Family and classes at 6 stations and bar graphs, pie charts, line graphs were prepared using Excel 2007.

Different diversity indices were calculated using the following formulas:

3.4.1. Family Richness (R)

The family richness or species richness of any ecosystem was built on the numbers of Family or number of species found in an area or ecological environment.

3.4.2. Shannon diversity index (H)

The collected data were pooled together to estimate the Shannon index to determine species diversity and species evenness. Shannon-wiener index (H) was calculated following the equations:

$$H = - \sum [(P_i) \times \ln (P_i)]$$

Where ,

$$P_i = n/N$$

n = Number of individuals of each species i

N = Total number of individuals of all species

[9]

3.4.3 Evenness (E)

Evenness of family was calculated by the following formula:

$$E = H/\ln(R)$$

Here,

H = Shannon-wiener index

R = Species richness [9]

4. Result and Discussion

Polychaeta, Malacostraca, Clitellata, Bivalvia, Turbellaria, Chromadorea, Pleistoannelida, Anopla.

Palaeonemertea, Oligochaeta, Malacostraca, and Enteropneusta . Gastropoda, Anthozoa, and Holothuroidae were identified in a total of 6 stations (Table 2) , that was 15 classes of Macrobenthos. 8 types of the phylum were identified (Table 3) , with Annelida having the highest abundance of macrobenthos in all stations. The abundance of the macrobenthos family at the Salimpur coast and the abundance of the macrobenthos family at Shapuree Island were identified, where the family Nereididae was one of the most numerous species. Of all classes, the maximum number of benthos abundance was polychaeta at all stations. On the Salimpur coast, the polychaeta class benthos abundance was 17%, 10%, and 13% at all 3 stations, and 11%, 26%, and 23% at stations 1, 2, and 3 at Shapuree Island. On the Salimpur coast, stations 1, 2, and 3 had 10, 8, and 6 types of macro benthos diversity, while 7, 9, and 10 had stations 1, 2, and 3, respectively on Shapuree Island, according to their macro benthos family and the abundance of the macrobenthos family at Shapuree Island (Table 1) were identified, where the family Nereididae was one of the most numerous species. Of all classes, the maximum number of benthos abundance was polychaeta at all stations.

Table 1: Macrobenthos abundance at Salimpur coast and Shapuree Island according to family

Family	Salimpur Coast			Shapuree Island		
	Station 1	Station 2	Station 3	Station 1	Station 2	Station 3
Arenicolidae	2	0	0	0	0	0
Sabellidae	3	0	0	0	0	0
Lumbrineridae	1	0	0	0	0	0
Gorgonorhynchidae	2	0	0	0	0	0
Cephalotrichidae	2	0	0	0	0	0
Lumbricidae	4	0	0	0	0	0
Nephtyidae	3	0	0	2	0	0

Nereididae	5	0	6	10	5	8
Eusiridae	3	0	0	0	0	0
Phyllodocidae	2	0	0	0	1	0
Capitellidae	0	1	5	0	0	3
Edwardsioidea	0	3	0	0	0	0
Paraonidae	0	2	0	0	0	0
Orbiniidae	0	1	0	0	0	0
Prorhynchida	0	5	0	0	0	0
Aspidosiphonidae	0	3	0	0	0	0
Piscicolidae	0	2	0	0	0	0
Myriotrochidae	0	1	0	0	0	0
Spionidae	0	0	1	2	2	2
Dironidae	0	0	4	0	0	0
Maldanidae	0	0	2	0	0	0
Serpulidae	0	0	3	0	0	0
Syllidae	0	0	0	1	0	0
Eunicidae	0	0	0	3	0	0
Enoturbellidae	0	0	0	3	0	0
Pilargidae	0	0	0	1	0	0
Trochochaetidae	0	0	0	0	0	0
Hesionidae	0	0	0	0	3	0
Atyidae	0	0	0	0	2	0
Naididae	0	0	0	0	6	0
Magelonidae	0	0	0	0	3	0
Glyceridae	0	0	0	0	4	0
Xyalidae	0	0	0	0	0	2
Polychaete	0	0	0	0	0	1
Xenoturbellidae	0	0	0	0	0	2
Lumbriculidae	0	0	0	0	0	2
Maldanidae	0	0	0	0	0	6
Veneridae	0	0	0	0	0	4
Melinnidae	0	0	0	0	0	3

Table 2: Class wise total abundance of macro benthos in all stations at Salimpur coast and Shapuree Island

Class	Number of benthos abundance	Percentage
Malacostraca	2	1.35%
Clitellata	2	1.35%
Bivalvia	4	2.70%
Turbellaria	2	1.35%
Chromadorea	2	1.35%
Pleistoannelida	34	22.97%
Anopla	2	1.35%
Palaeonemertea	2	1.35%
Oligochaeta	4	2.70%
Malacostraca	3	2.03%
Enteropneusta	1	0.68%
Gastropoda	4	2.70%
Anthozoa	3	2.03%
Holothuroidae	1	0.68%
Polychaeta	82	55.41%

Table 3: Phylum wise abundance of macro benthos in all stations at Salimpur coast and Shapuree Island

Phylum	Total number of benthos	Percentage
Nemertea	4	2.70%
Arthropoda	6	4.05%
Echinodermata	1	0.68%
Mollusca	4	2.07%
Hemichordata	2	1.35%
Xenacoelomorpha	3	2.03%
Nematoda	3	2.03%
Annelida	125	84.46%

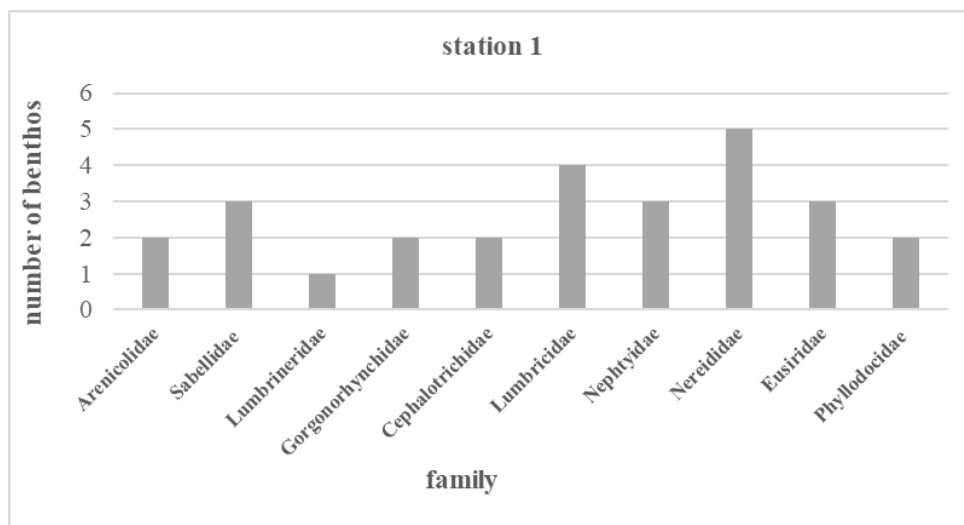


Figure 5: Number of benthos according to family at Salimpur coast station 1

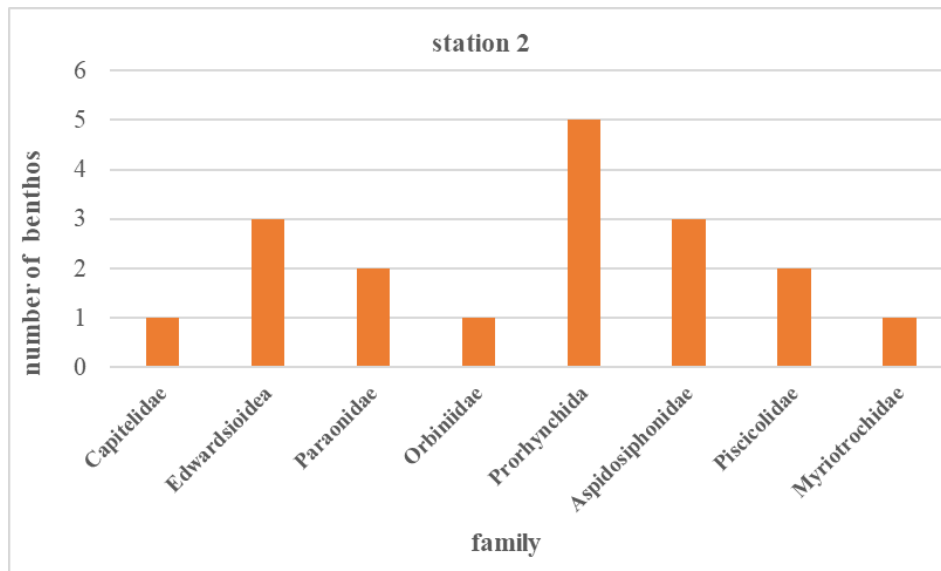


Figure 6: Number of benthos according to family at Shapuree Island station 2

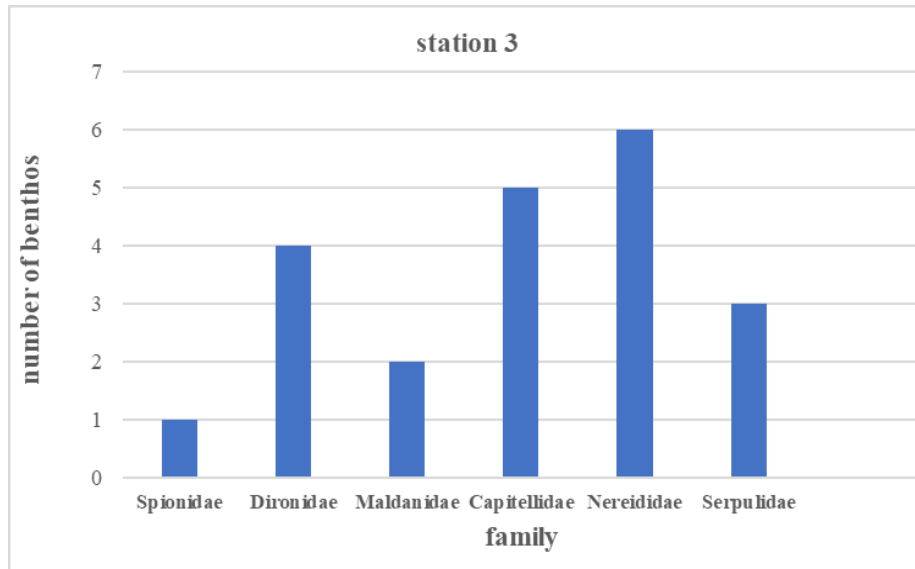


Figure 7: number of benthos according to family at Salimpur coast station 3



Figure 8: Variation in 3 stations of benthos abundance at Salimpur coast

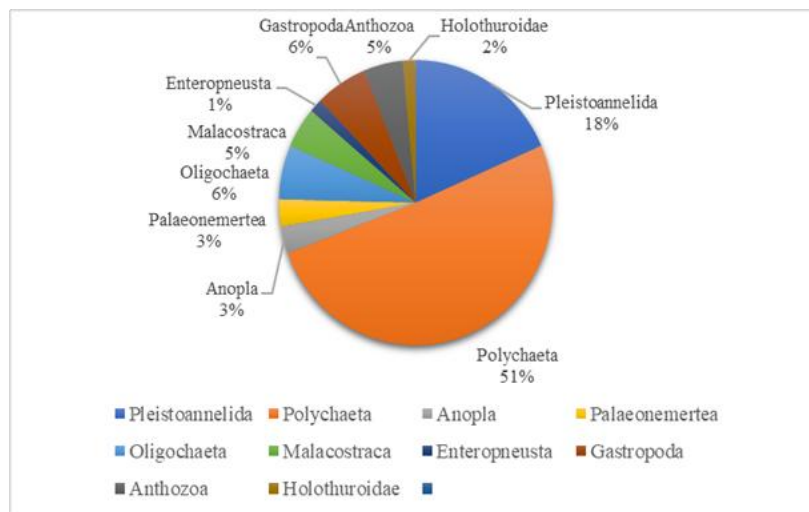


Figure 9 : Percentage of benthos abundance according to class at Salimpur 3 stations

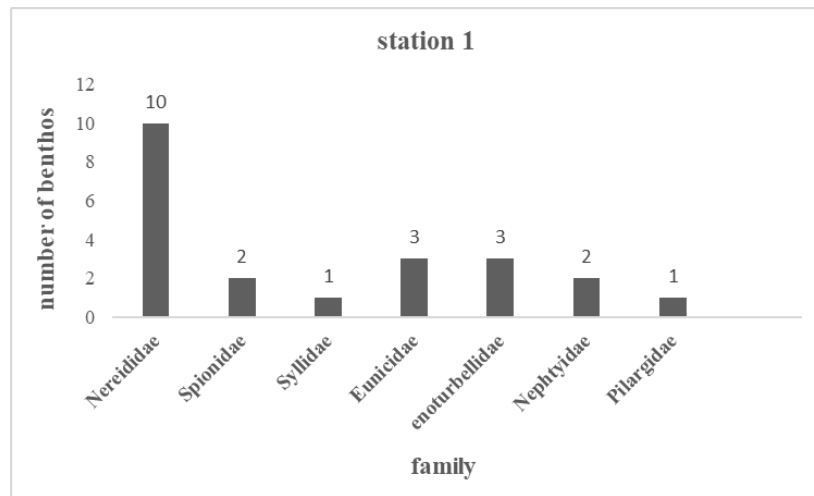


Figure 10: Number of benthos according to family at Shapuree Island station 1

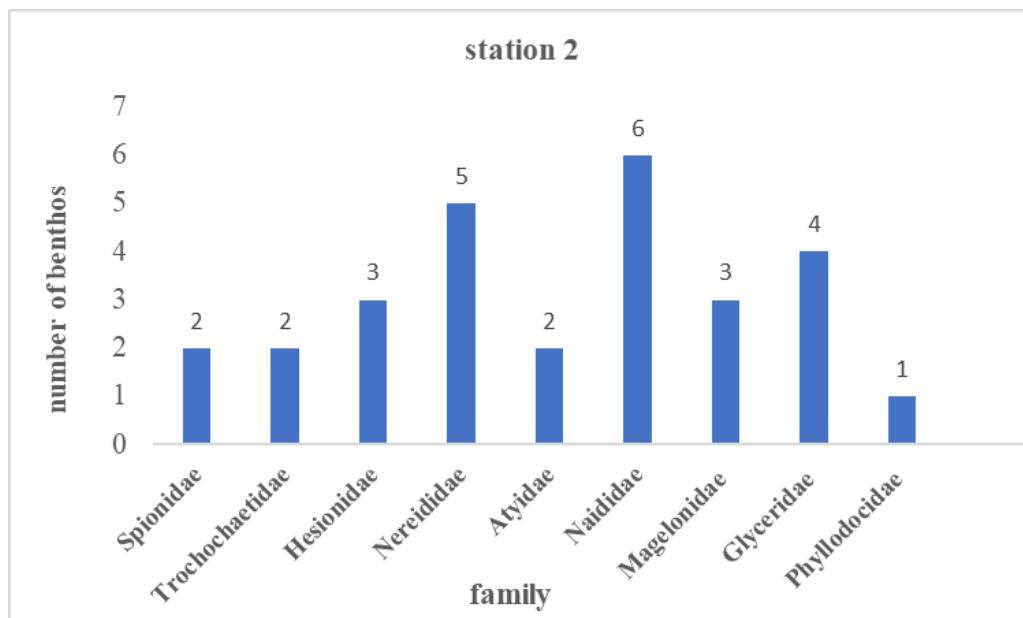


Figure 11: Number of benthos according to family at Shapuree Island station 2

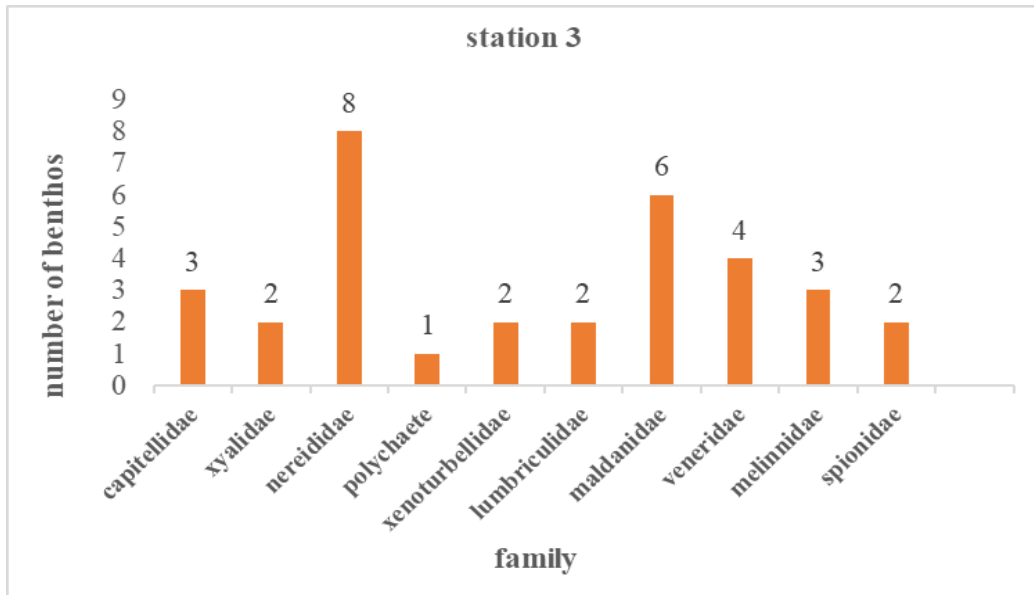


Figure 12: Number of benthos according to family at Shapuree Island station 3



Figure 13: Variation in 3 stations of benthos abundance at Shapuree Island

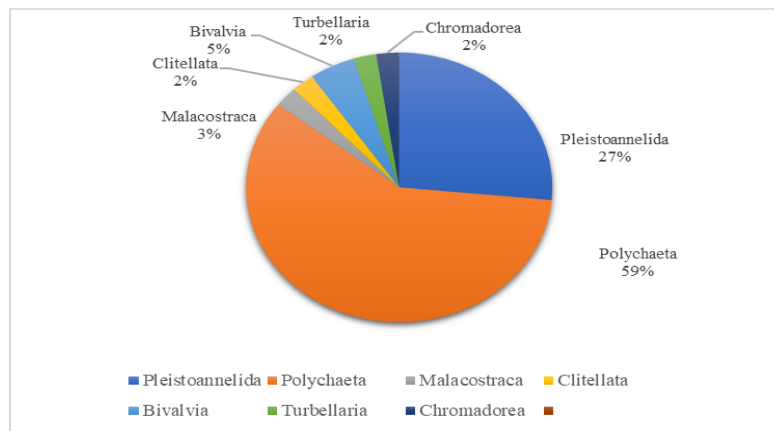


Figure 14 : Benthos abundance according to class at Shapuree Island 3 stations

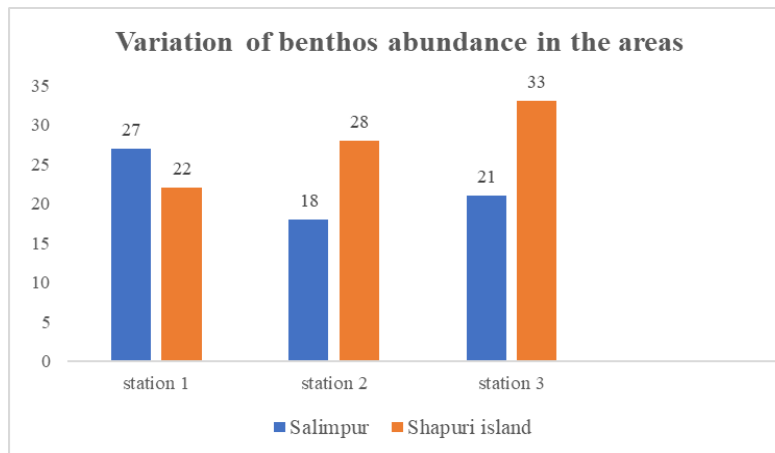


Figure 15: Variation of total benthos abundance at Salimpur coast and Shapuree Island stations.

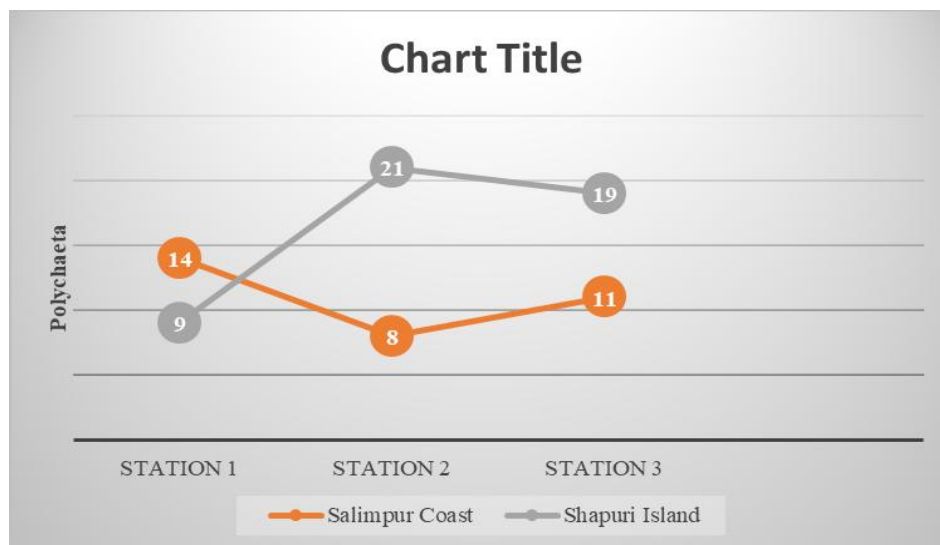


Figure 16: Maximum number of polychaeta class benthos and variation at all stations

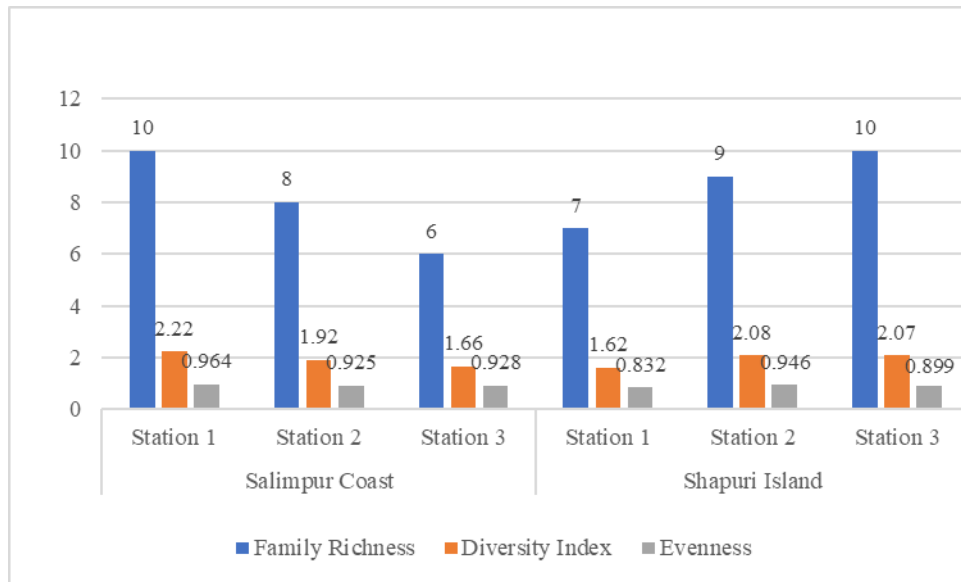


Figure 17 : 2 areas 3 stations family richness , diversity index, evenness variation.

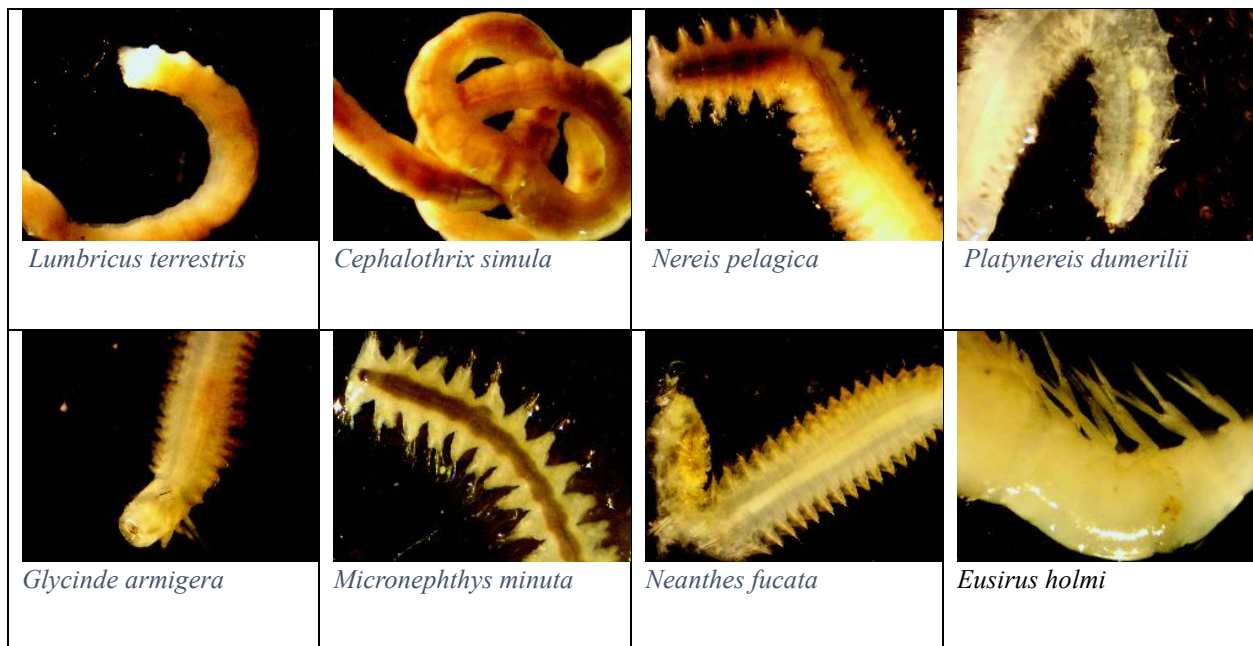


Figure 18 : Some macrobenthos species pictures from station 1 at Salimpur Coast.



Figure 19 : Some macrobenthos species pictures from station 2 at Salimpur Coast, . *Notomastus latericeus*, *Edwardsioidea andres* , *Levinsenia mesnil* , *Orbiniidae* Hartman, *Prorhynchus fontinalis*, *Lithacrosiphon cristatus*, *Myzobdella lugubris*, *Myriotrochus eurycyclus* (from left to right)



Figure 20 : Some macrobenthos species pictures from station 3 at Salimpur Coast. *Saccoglossus pusillus* ,*Dirona picta*, *Praxillella antarctica* ,*Capitella capitata*, *Platynereis massiliensis*, *Platynereis festiva*, *Hydroides tubeworm*, *Boccardiella hamata* (from left to right).



Figure 21 : Some macrobenthos species pictures from station 1 at Shapuree Island. *Alitta scccinea* , *Eumida sanguinea*, *Prionospio aucklandica* , *Marphysa sanguinea* , *Neanthes fucata* , *Xenoturbella japonica* , *Micronephthys minuta* , *Pilargis maculate* , *Syllis gracillis*. (from left to right)



Figure 22 : Some macrobenthos species pictures from station 2 at Shapuree Island. *Aurospio banyulensis* , *Sirsoe munki*, *Alitta succinea* , *Prionospio Malmgren* , *Trochochaeta levinsen* , *Neanthes augeneri*, *Caridina multidentate*, *Namanereis chamberlin* , *Amphichaeta leydigii* , *Magelona pacifica* , *Neanthes nubila* , *Eulalia ornata* . (from left to right)



Figure 23 : Some macrobenthos species pictures from station 3 at Shapuree Island. *Capitella capitata* , *Amphimonhystera molloyensis* , *Alitta virens* , *Glyceridae grube* , *Xenoturbella bocki* , *Lumbriculus olgae* , *Nicomache lumbricalis* , *Laeonereis culveri* , *Praxillella pacific* , *Marcia marmorata* , *Melinna elisabethae* , *Prionospio malmgren*. (from left to right)

The comparison of macrobenthos variation between Salimpur Coast and Shapuree Island, as reflected in the three stations for each location, reveals intriguing insights into the abundance and diversity of marine life. Examining family richness, Salimpur Coast's Station 1 stands out with 10 families, surpassing the respective counts at Stations 2 and 3. Shapuree Island, on the other hand, displays an increasing trend in family richness from Station 1 to Station 3. The diversity index further underscores the contrasting ecological dynamics, with Salimpur Coast showcasing a gradual decline in diversity from Station 1 to Station 3, while Shapuree Island exhibits a more variable pattern. Notably, Station 2 on Shapuree Island records the highest diversity index (Figure 17). Evenness values highlight the distribution of species within each station, with Salimpur Coast generally displaying higher evenness compared to Shapuree Island. The comparison of macrobenthos variation between Salimpur Coast and Shapuree Island, as reflected in the three stations for each location, reveals intriguing insights into the abundance and diversity of marine life. Examining family richness, Salimpur Coast's Station 1 stands out with 10 families, surpassing the respective counts at Stations 2 and 3. Shapuree Island, on the other hand, displays a decreasing trend in family richness from Station 1 to Station 3. The diversity index further underscores the contrasting ecological dynamics, with Salimpur Coast showcasing a gradual decline in diversity from Station 1 to Station 3, while Shapuree Island exhibits a more variable pattern. Notably, Station 2 on Shapuree Island records the highest diversity index. Evenness values highlight the distribution of species within each station, with Salimpur Coast generally displaying higher evenness compared to Shapuree Island. A varied distribution of benthic families is shown in the macrobenthos abundance data (Figure 8) from Salimpur Coast Stations 1, 2, and 3 in Chittagong, which reflects the various ecological conditions across these stations. While Station 2 emphasizes the predominance of Prorhynchida with 5 members, indicating a significant role in sediment dynamics, Station 1 displays a balanced composition with Nereididae dominating at 5 members. However, Station 3 highlights the flexibility of Nereididae, which has six members, and the significant presence of Capitellidae, which has five. This diversity highlights the Salimpur coast ecological complexity, as each station supports a unique benthic community that is shaped by regional environmental conditions. Regarding variation, the observed variations in family abundance between the stations could indicate different environmental conditions like different types of sediment, different water depths, and different nutrient availability. For example, the predominance of Prorhynchida at Station 2 might suggest particular preferences for sediment, whereas the Nereididae are prevalent in all stations, indicating their environmental adaptability.

The type of sediment present at Salimpur Coast is crucial in understanding the distribution of these benthic families. Arenicolidae, Lumbricidae, and Nereididae, which dominate at Station 1, are often associated with sandy substrates. In contrast, Prorhynchida, Edwardsioidea, and Aspidosiphonidae at Station 2 are indicative of muddy or sediment-rich environments. Station 3, with Capitellidae and Nereididae as major players, may signify a mix of sandy and muddy substrates.

The macrobenthos abundance data from Shapuree Island, as reflected in Table 2, unveils a diverse array of benthic families across the three stations. Station 1 is characterized by a mix of families, with Nereididae standing out at 10 members, highlighting their prominence in this habitat. Eunicidae, Enoturbellidae, and Nephtyidae contribute to the diversity, indicating a rich and varied

macrobenthic community. Station 2 exhibits a broader spectrum, showcasing the presence of families such as Nereididae, Atyidae, Naididae, and Glyceridae, each with notable representation. Capitellidae and Xyalidae add to the diversity, while Station 3 presents a distinct composition, with Maldanidae dominating at 6 members, emphasizing their significant role in the benthic ecosystem. Spionidae, Veneridae, and Melinnidae also contribute to the diversity at this station. Additionally, the presence of Syllidae, Eunicidae, and Nephtyidae underscores the ecological diversity of this station. Station 2 stands out for its notable representation of Naididae, Glyceridae, and Atyidae, suggesting a mix of sediment types, potentially including sandy and muddy substrates. The variety in families such as Magelonidae and Phyllodocidae further highlights the ecological complexity at this station. Station 3, characterized by the dominance of Maldanidae and Veneridae, may indicate a preference for sandy substrates. The presence of diverse families like Xenoturbellidae and Lumbriculidae adds to the habitat diversity.

Table 2 provided a class-wise breakdown of the total abundance of macrobenthos across all stations at Salimpur Coast and Shapuree Island. Polychaeta dominates the overall composition, constituting 55.41% of the total abundance, and emerges as the most abundant class across both locations, securing the top rank. Pleistoannelida follows closely at 22.97%, securing the second rank. Bivalvia, Oligochaeta, and Gastropoda each contribute 2.70%, while Anthozoa and Malacostraca account for 2.03%. 1.35% for Clitellata, Turbellaria, Chromadorea, Anopla, Palaeonemertea where Enteropneusta, and Holothuroidae each represent 0.68 % of the total abundance, ranking lower in their respective classes.

The dominance of Polychaeta, characterized by versatile and adaptable worm-like organisms, suggests a resilient and thriving benthic community at both Salimpur Coast and Shapuree Island. Polychaeta class benthos can absorb high temperature by using their hydral vents [25].Pleistoannelida, comprising various marine worms, also contributes significantly to the overall abundance, indicating a diverse ecosystem. The presence of Bivalvia, Oligochaeta, and Gastropoda highlights the varied habitat preferences and ecological roles of different macrobenthic classes. The relatively lower abundance of classes like Clitellata, Turbellaria, Chromadorea, Anopla, Palaeonemertea, Enteropneusta, and Holothuroidae suggests their lesser contribution to the overall benthic community.

Table 3 provided a phylum-wise overview of the macrobenthos abundance across all stations at Salimpur Coast and Shapuree Island. Annelida emerges as the dominant phylum, constituting 84.46% of the total abundance and securing the top rank. Arthropoda and Mollusca contribute 4.05% and 2.07% each, where the second rank of Arthropoda, while Nemertea and Xenacoelomorpha each represent 2.70% and 2.3%. Nematoda accounts for 2.03%, and Echinodermata and Hemichordata each contribute 0.68% and 2.03%. The overwhelming dominance of Annelida, which includes segmented worms like polychaetes, underscores their ecological significance in both Salimpur Coast and Shapuree Island ecosystems. [7] they identified the benthos species and found crustacea was the most dominated group of species. [26] they were identified their species by five major taxa where polychaeta was larger number of species group .This high abundance suggests a strong representation of Annelids in various habitats and environmental conditions, indicating adaptability and resilience.

In Salimpur Coast, Station 1 records 27, Station 2 has 18, and Station 3 has 21 benthic organisms. In Shapuri Island, Station 1 has 22, Station 2 has 28, and Station 3 has 33. These variations in abundance across stations and locations provide valuable insights into the distribution and dynamics of benthic communities in the respective areas.

The higher total abundance in Shapuree Island, especially at Stations 2 and 3, may suggest a more diverse or favorable habitat for benthic organisms compared to Salimpur Coast. The differences in abundance between stations within each location indicate potential variations in environmental factors such as sediment type, water quality, or ecological interactions. For example, the higher abundance at Station 3 in both Salimpur Coast and Shapuree Island may suggest a particularly suitable environment for benthos, possibly due to specific sediment characteristics or ecological conditions.

Polychaeta abundance at different stations on the Salimpur Coast and Shapuree Island. Notably, Station 2 on Shapuri Island exhibits a higher abundance of polychaeta, suggesting it provides favorable conditions for these segmented worms. Conversely, lower abundances at Station 1 on Shapuri Island and Station 2 on Salimpur Coast indicate potential differences in habitat suitability or environmental factors influencing Polychaeta presence.

Hence, some species occur in sandy, muddy, or sandy and muddy areas along the Salimpur coast. Besides, the species on Shapuree Island are either sandy or sandy and muddy. 68 species of benthos were found at Salimpur 3 station and 83 species at Shapuree Dwip 3 station. Variation of this abundance for sediment composition, environmental conditions, and species adaptability since we collected samples from the same amount of area, which was 1 m for 1 station.

[15] polychaeta was major macrofuna in the Hatiya and Nijhum Dwip. In our study area, maximum diversity index had 2.22 at Salimpur coast's station 1, minimum diversity index had 1.62 at Shapuree Island's station 1. [27] observed minimum value 1.22 and maximum value 2.70. [28] observed maximum value from 1.49 and minimum value from 1.20 [29] 1.97 of maximum value and 1.24 of minimum value from Shannon diversity index. Salimpur coast areas had 27,18,21 respectively station 1,2,3 for individuals per square meter and 22,28,31 individuals per square meter species had station 1,2,3 respectively at Shapuree Island [28] macrobenthos abundance at Mouri river for individuals per square meter 630-1040. [30] the average Ganga river macrobenthic abundance was 119-4055 individuals per square meter. Figure 17 showed the macrobenthos abundance difference between Salimpur coast and Shapuree Island according to diversity index, evenness, richness.

5. Conclusion

The study focused on evaluating the abundance and diversity of macrobenthos in the Salimpur coast and Shapuree Island area, specifically at three stations each. The investigation included the identification of 39 families, totaling 148 species across all stations. Notably, Salimpur Station 1 and Shapuree Station 3 stood out with 10 different families and the highest species diversity. The family Nereididae emerged as the most abundant species, prevalent in all stations except Salimpur Coast Station 2. The analysis also extended to phylum and class, revealing that Annelida, constituting 84.46% of the identified species, was the dominant phylum. Within Annelida, Polychaeta held the highest percentage, accounting for 55.41% of the total species. This comprehensive assessment sheds light on the intricate ecological dynamics of the macrobenthic community in the Salimpur coast and Shapuree Island area. The variation in family distribution and the dominance of certain species, such as Nereididae and Polychaeta, contribute valuable insights into the biodiversity and ecological composition of these coastal regions.

6. Reference

- [1] D. S. a. E. M. McLusky, "The estuarine ecosystem: ecology, threats and management.," Oxford University Press, vol. Oxford , no. third edition , p. 214, 2004.
- [2] S. G. M. B. M. A. V. C. A. P. G. O. d. F. L. C. & F. M. Carvalho, " The use of the marine biotic index AMBI in the assessment of the ecological status of the Óbidos lagoon (Portugal)," Marine Pollution Bulletin, p. 1414–1424, 2006.
- [3] M. A. K. a. P. Shrivastava, "Diversity, Abundance and Seasonal Variation of Benthic Macroinvertebrates in Kolar Reservoir, Central India," Bulletin of Environment, Pharmacology and Life Sciences, pp. 141-149, 2021.
- [4] H. Caspers, "Dales, R. Phillips: Annelids. London: Hutchinson & Co. (Publ.) Ltd. 1963. 200 pp. = Hutchinson University Library. 15 s," Internationale Revue der gesamten Hydrobiologie und Hydrographie, p. 469–469, 1963.
- [5] ". Regional Aquatics Monitoring Program (RAMP)," RAMP Online, 2017.
- [6] B. CE, "Water quality in warm water fish ponds," University, Agricultural Experiment Station, Auburn, , pp. 9-44, 1979.
- [7] a. M. B. H. M. M. I. M. M. B. A. A.-A. M. M. Billah, "Diversity and abundance of Macrobenthos in a subtropical estuary, Bangladesh," Bangladesh. Species, pp. 140-150, 2018.
- [8] Y. A. Habib S, "Benthic macroinvertebrate community of Yousmarg streams (Doodganga stream and Khanshah Mansbah canal) in Kashmir Himalaya, India," J. Ecol. Natur. Environ, pp. 280-289, 2012.
- [9] M. M. H. A. S. M. S. M. K. A. S. R. M. H. R. M. a. M. I. Khan, "Macrobenthic Faunal Abundance, Distribution and Diversity in the Bakkhali River, East Coast of Bangladesh," The Dhaka University Journal of Earth and Environmental Sciences, pp. 48-55, 2021.
- [10] M. B. H. a. M. M. M. HOSSAIN, "A PRELIMINARY STUDY ON INTERTIDAL MACROBENTHOS OF CHITTAGONG COAST IN BANGLADESH," Int. J. Ani. Fish. Sci., pp. 170-175, 2009.
- [11] M. V. S. K. A. a. W. P, "Biodiversity and Abundance of Benthic Macro-invertebrates Community of River Tawi in Vicinity of Udhampur City (J&K) INDIA," International Research Journal of Environment Sciences, pp. 17-24, 2013.

- [12] A. N. K. D. M. M. M. R. M. A. a. H. M. A. Khan, ". Diversity, Distribution and Abundance of Benthos in Mouri River, Khulna, Bangladesh.," *Int. J. Sustain. Crop Prod.*, pp. 19-23 , 2007.
- [13] M. A. K. a. P. Shrivastava, "Diversity, Abundance and Seasonal Variation of Benthic Macroinvertebrates in Kolar Reservoir, Central India," *Bulletin of Environment, Pharmacology and Life Sciences*, pp. 141-149, 2021.
- [14] G. W. S. L. D. M. M. A. G. M. M. A. a. Z. C. Cob, "Macrobenthos Composition, Distribution and Abundance within Sungai Pulai Estuary, Johor, Malaysia," *RESEARCH ARTICLE*, p. 591–596 , 2014.
- [15] M. A. M. B. H. M. S. M. A. a. A. Azam, "Occurrence and Abundance of Macrobenthos of Hatiya and Nijhum Dweep Islands, Bangladesh," *Middle-East Journal of Scientific Research*, pp. 184-188, 2012.
- [16] M. I. M. I. C. O. S. S. F. M. N. a. A. S. BATUBARA, "The abundance and diversity of benthic community in Krueng Cut estuary, Banda Aceh, Indonesia," *BIHAREAN BIOLOGIST* , pp. 85-89 , 2020.
- [17] a. J. A.-O. A.D.I. George J.F.N. Abowei, "The Distribution, Abundance and Seasonality of Benthic Macro Invertebrate in Okpoka Creek Sediments, Niger Delta, Nigeria," *Research Journal of Applied Sciences Engineering and Technology*, pp. 11-18, 2010.
- [18] M. B. H. Md. Abu NOMAN MAMUNUR Rashid M SHAHANUL Islam, "Spatial and seasonal distribution of intertidal macrobenthos with their biomass and functional feeding guilds in the Naf River estuary, Bangladesh," *Journal of Oceanology and Limnology*, pp. 1010-1023, 2019.
- [19] A. S. M. . D. V. Desai, "Distribution and abundance of macrobenthic polychaetes along the South Indian coast," *Environ Monit Assess*, p. 423–436, 2011.
- [20] J. S. P. K. Behera, "Abundance & distribution of aquatic benthic macro-invertebrate families of river Ganga and correlation with environmental parameters," *Environ Monit Assess*, p. 546, 2022.
- [21] Y. I. •. T. T. •. Y. A. T. Takahashi, "Annual changes in macrobenthos abundance in Funka Bay, Japan," *The Japanese Society of Fisheries Science*, p. 647–659, 2012.
- [22] B. I. S. a. S. Y. Gagaev, "Unusual Abundance of Macrobenthos and Biological Invasions in the Chukchi Sea," *ISSN 1063-0740, Russian Journal of Marine Biology*, p. 355–364, 2007.
- [23] N. A. S. K. A. D. a. V. S. K.K. Sharma, "Biodiversity and Abundance of Benthic Macro Invertebrates Community Of Datte-Da-Talab Pond, Birpur (J&K) India," *International Multidisciplinary Research Journa*, pp. 13-17, 2013.

- [24] A. A. M. A. K. A. a. S. M. S. M. K. Abu Hena, "Distribution of Aquatic Macrophytes in the Coastal Area of Salimpur, Chittagong, Bangladesh," CMU. J. Nat. Sci., pp. 273-279, 2010.
- [25] https://en.wikipedia.org/wiki/Polychaete#cite_note-2.
- [26] K. S. S. M. I. J. I. M. A. M. Abu Hena MK, "Composition of Macrobenthos in the Bakkhali Channel System, Cox's Bazar with Notes on the Soil parameter," Pakistan J. Biol. Sci, pp. 641-646, 2012.
- [27] H. S. G. R. M. M. Montajami S, "Investigation of Some Physicochemical Characteristics of Farobroman River Water by Using Benthic Macro invertebrates as Biological Indicator.," World J. Fish Mar. Sci, pp. 645-650., 2012.
- [28] K. D. M. M. R. M. H. M. Khan AN, "Diversity, Distribution and Abundance of benthos in Mouri River, Khulna, Bangladesh.," Int. J. Sustain. Crop Prod, pp. 73-87, 2007.
- [29] A. S. A. M. R. S. Hossain MB, "Analyses of Macrobenthos of Hatiya and NijhumDweep Islands at higher Taxonomic Resolution," J. Fish. Aquat. Sci, pp. 526-534, 2013.
- [30] M. AK, "Organizational responses to crisis: The centrality of trust. In R. M. Kramer & T. R. Tyler (Eds.)," Trust in Organizations: Frontiers of Theory and Research, pp. 261-287, 1996.
- [32] "Water quality in warm water fish ponds," Auburn University, Agricultural Experiment Station, Auburn,Alabama, USA,, pp. 9-44, 1979.